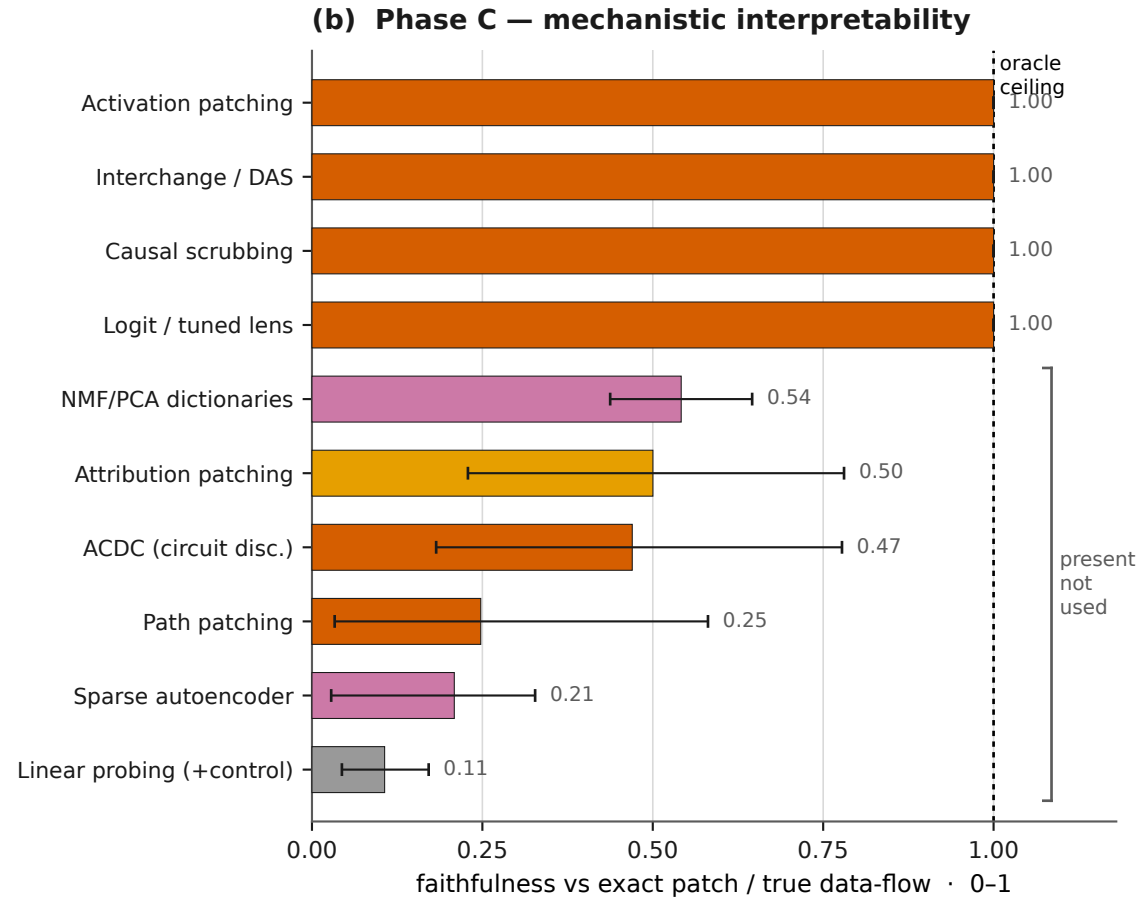
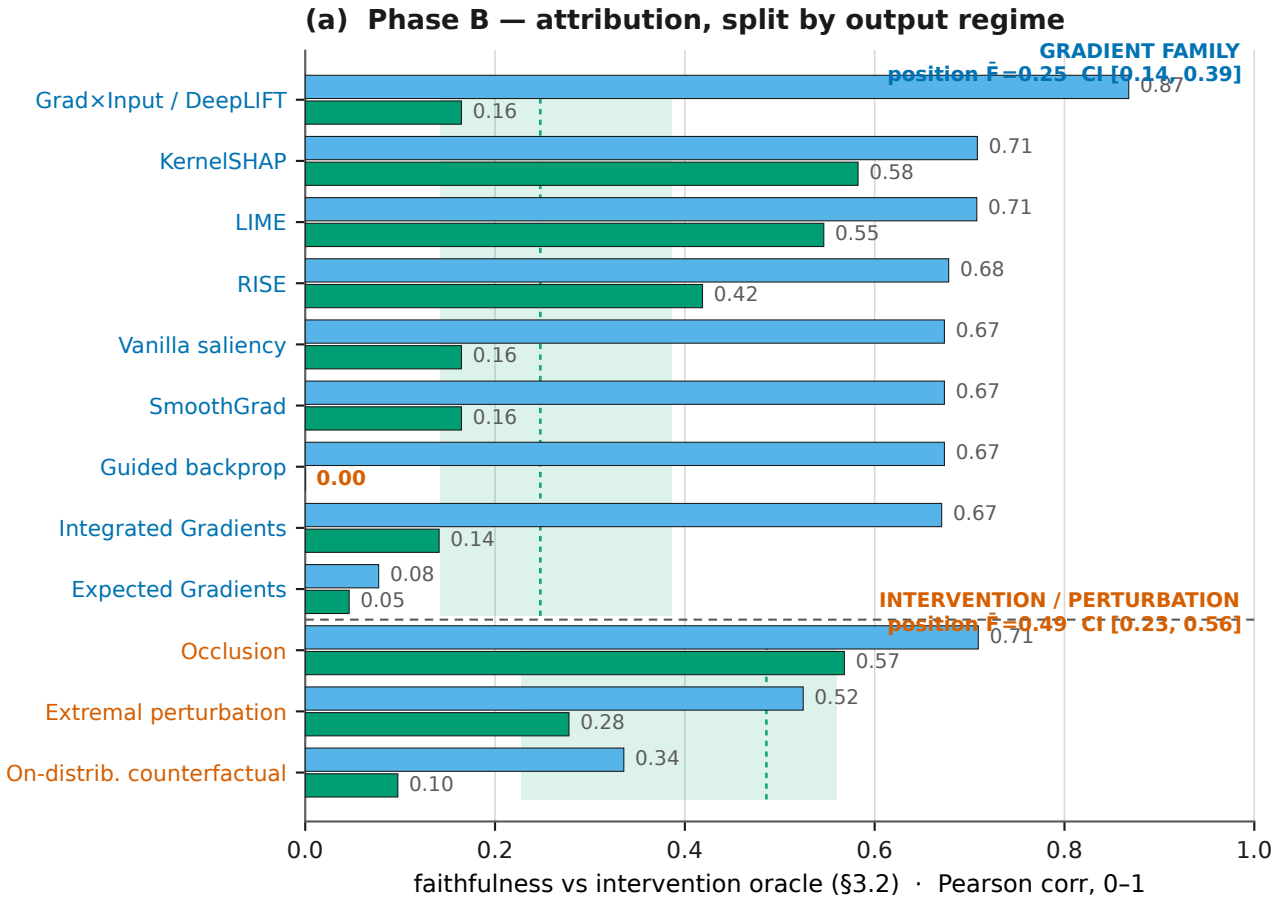


Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

The 22 Phase-B + Phase-C method rows (a subset of the 30 methods + the oracle = 31-row leaderboard; the cross-tradition scatter is Fig. 2).

HEADLINE — across ALL output regimes, intervention/causal methods are more faithful to the true mechanism than gradient/correlational ones. Causal/intervention $\bar{F}=0.664$ (95% CI [0.599, 0.741]) vs gradient/correlational $\bar{F}=0.367$ (95% CI [0.324, 0.411]) — a 0.297 gap, 95% CI [0.232, 0.376] (excludes 0: robust). On the POSITION/INDEX regime alone (discrete sprite-position outputs), the gap is DIRECTIONAL but not significant at 6 games: causal 0.486 (n=4) vs gradient 0.248 (n=9), gap 0.238, 95% CI [-0.049, 0.315] (includes 0).



(a) output regime
content output position / index output

(b) Phase-C category + uncertainty
causal (exact / partial) gradient approximation dim-reduction / SAE (present-not-used) probing (present-not-used) — 95% CI (bootstrap over)

Data: committed records (leaderboard + bootstrap CIs + faithful-demo); 6 core games, 30-frame horizon; pure read — no experiment re-run. Exact paths & per-number provenance in the Supplement.